

Disproof of TMC Conjecture 00000000433: the Case $\mathbf{B}_2$ of the Denominator-Terms Conjecture

TLMC falsification pipeline submission

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Abstract

Conjecture 00000000433 asserts that if $T(\text{type})$ denotes the number of monomial terms in Macdonald's denominator identity, then $T(\mathbf{B}_n) = T(C_n) = n$, $T(D_n) = 2n$, $T(G_2) = 2$, $T(F_4) = 4$, $T(E_6) = 6$, $T(E_7) = 7$, $T(E_8) = 8$, and that the number of terms always equals the multiplicity of the exponent 1. We disprove the conjecture at $\mathbf{B}_2$: a direct computation of the Weyl-group orbit of ρ shows the denominator identity of $\mathbf{B}_2$ has exactly 8 monomial terms, i.e. $T(\mathbf{B}_2) = 8 \neq 2 = n$. The failure is general: $T(\text{type}) = |W(\text{type})|$ for every root system in the list, so all displayed values are false.

1 The conjecture

Macdonald's denominator identity expresses the denominator of an affine Lie algebra as a linear combination of monomials; let $T(\text{type})$ be its number of terms. The conjecture claims $T(\mathbf{B}_n) = T(C_n) = n$, $T(D_n) = 2n$, $T(G_2) = 2$, $T(F_4) = 4$, $T(E_6) = 6$, $T(E_7) = 7$, $T(E_8) = 8$, and that the number of terms always equals the multiplicity of exponent 1.

2 The attack: $\mathbf{B}_2$

The finite-type right-hand side of Macdonald's denominator identity is the Weyl denominator sum

$$\prod_{\alpha \in \Delta^+} (1 - e^{-\alpha}) = \sum_{w \in W} \varepsilon(w) e^{w(\rho) - \rho}, \quad \rho = \frac{1}{2} \sum_{\alpha \in \Delta^+} \alpha.$$

Coefficients are ± 1 , so the number of monomial terms equals the number of *distinct* values of $w(\rho) - \rho$ as w ranges over W .

Realize $\mathbf{B}_2$ orthonormally: $\Delta = \{\pm e_1, \pm e_2, \pm e_1 \pm e_2\}$, $\Delta^+ = \{e_1, e_2, e_1 - e_2, e_1 + e_2\}$, so

$$\rho = \frac{1}{2}(e_1 + e_2 + (e_1 - e_2) + (e_1 + e_2)) = \left(\frac{3}{2}, \frac{1}{2}\right),$$

and $W(\mathbf{B}_2)$ is the group of the 8 signed coordinate permutations $x \mapsto (\pm x_{\sigma(1)}, \pm x_{\sigma(2)})$.

All eight values are pairwise distinct. Indeed ρ is strictly dominant, hence has trivial stabilizer, so $w(\rho) = w'(\rho)$ implies $w = w'$; no two terms of the sum share a monomial and no cancellation can occur. Therefore

$$T(\mathbf{B}_2) = 8 \neq 2 = n,$$

which falsifies the conjecture. As a byproduct, the companion clause fails too: the exponents of $\mathbf{B}_2$ are $\{1, 3\}$, so the multiplicity of exponent 1 is $1 \neq 8 = T(\mathbf{B}_2)$.

$w \in W(\mathbf{B}_2)$	$w(\rho) - \rho$
$x \mapsto (x_1, x_2)$	$(0, 0)$
$x \mapsto (-x_1, x_2)$	$(-3, 0)$
$x \mapsto (x_1, -x_2)$	$(0, -1)$
$x \mapsto (-x_1, -x_2)$	$(-3, -1)$
$x \mapsto (x_2, x_1)$	$(-1, 1)$
$x \mapsto (-x_2, x_1)$	$(-2, 1)$
$x \mapsto (x_2, -x_1)$	$(-1, -2)$
$x \mapsto (-x_2, -x_1)$	$(-2, -2)$

Table 1: The eight exponents $w(\rho) - \rho$ for $\mathbf{B}_2$.

Independent cross-check. In simple-root coordinates (α_1 short; Cartan matrix $\begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix}$); reflections $s_1(a, b) = (-a + 2b, b)$, $s_2(a, b) = (a, a - b)$; positive roots $\alpha_1, \alpha_2, \alpha_1 + \alpha_2, 2\alpha_1 + \alpha_2$; take the integral multiple $2\rho = (4, 3)$, harmless since scaling by 2 is a bijection), the eight values $w(2\rho) - 2\rho$ are

$$(0, 0), (-2, 0), (0, -2), (-6, -2), (-2, -4), (-8, -4), (-6, -6), (-8, -6),$$

again pairwise distinct, with sanity checks $w_0(2\rho) = -2\rho$ and $\sum_w (w(2\rho) - 2\rho) = -8(4, 3)$. This also reproduces the pipeline verdict list entry-for-entry in the orthonormal realization.

3 Generality of the failure

Since ρ is strictly dominant for every root system, its stabilizer is trivial and the $|W|$ exponents $w(\rho) - \rho$ are pairwise distinct; hence $T(\text{type}) = |W(\text{type})|$ throughout:

type	conjectured T	actual $T = W $
$\mathbf{B}_1 (= A_1 = C_1)$	1	2
$\mathbf{B}_n, C_n$	n	$2^n n!$
D_n	$2n$	$2^{n-1} n!$
G_2	2	12
F_4	4	1152
E_6	6	51840
E_7	7	2903040
E_8	8	696729600

Even $n = 1$ fails ($T(\mathbf{B}_1) = 2 \neq 1$), so there is no boundary case in which the conjecture holds.

4 Machine verification

The accompanying `reproduce.py` recomputes both realizations in exact rational arithmetic and asserts the table above entry-for-entry. The directory `lean4/` contains a core-Lean (no Mathlib) formalization: the eight exponents are produced by the eight explicit signed permutations, and their pairwise distinctness, the count 8, and the negation of $T(\mathbf{B}_2) = 2$ are proved by `rfl/decide` alone. Every theorem is axiom-free: `#print axioms` reports “does not depend on any axioms” for each, with no `sorry` and no `native_decide`.

5 Conclusion

Verdict: FALSE. Conjecture 00000000433 fails at B_2 : $T(B_2) = 8 \neq 2$, and in fact $T(\text{type}) = |W(\text{type})|$ for every root system named by the conjecture.